AEROSPACE PROPULSION

IV Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6AE09	PCC	3	0	0	3	40	60	100

UNIT-I **BACKGROUND AND REQUIREMENTS**

Introduction, Motion of bodies in space, parameters describing motion of bodies, frame of reference Impulse, force, universal law for gravitational force, motion in rotating frame of reference, pseudo- centrifugal force Orbits, orbit velocities, orbital period, geosynchronous and geostationary orbits, eccentricity and inclination, polar, sun-synchronous and other orbits Energy requirements for orbit, escape velocity

UNIT-II **NOZZLES AND ROCKETS**

NOZZLES: Rocket Nozzles:Expansion of gases from high pressure chamber, efflux velocity, classification, Performance losses in nozzles, Under-and Over- expansion nozzles (for very large pressure ratios),Performance loss in nozzles, mass flow rates and characteristic velocity role for interplanetary and deep space missions
ROCKETS: Introduction -Rocket principle and Tsiolkovsky Rocket equation Mass ratio of rocket, desirable parameters to achieve high velocities, propulsive efficiency Performance parameters of a rocket, staging and clustering

UNIT-III **SOLID PROPELLANT ROCKETS**

Solid propellant rockets: Introduction, classification Propellant grain configurations and its loading strategies: design of solid propellant rocket, Ignition of solid propellant rockets, ignition problems and solutions Characteristic burn times and action time. Liners, Insulators and inhibitors- Thrust developed by rocket , Net- and Gross- thrust, specific impulse- Calculation of heat of combustion, temperature, molecular mass and rocket performance parameters

UNIT-IV **LIQUID PROPELLANT ROCKETS AND CRYOGENICS**

LIQUIDPROPELLANT ROCKETS: Introduction to liquid propellant rockets, propellant feed systems, Gas Pressure feed system, Propellant properties, Liquid oxidizers and fuels
Cryogenics : Fundamentals of cryogenic s – components-Cryogenic fuels/propellant-Cryogenic safety aspects- Basic cryomodule design considerations- Next generation cryogenic engine

UNIT-V **COMBUSTION INSTABILITY IN ROCKETS AND ELECTRICAL ROCKETS**

COMBUSTION INSTABILITY IN ROCKETS: Combustion instability in rockets; illustration through examples, bulk and wave modes of instability, Modeling of bulk mode of instability in solid and liquid propellant rockets
ELECTRICAL ROCKETS
Introduction- electrical and magnetic fields, electro-thermal, arc-jet, electrostatic and electromagnetic thrusters

Text Books:

1. Ramamurthi, K., Rocket Propulsion, Macmillan (in press) 2009
2.Sutton, G. P. and Biblarj, O. Rocket propulsion elements, 7th Ed., New York: Wiley IntescienceePulications, 2001.

Reference Books:

1. Mechanics & Thermodynamics of Propulsion
2. M. Barrere, A. Jaumotte, B.J. Veubeke and J. Vanderkerckhove, Rocket Propulsion, Elsevier Publishing Company, Amsterdam, 1960 (remarks: no reference books available)

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Illustrate the basic concepts about motion of bodies in space and orbits
2. Demonstrate the rocket launching principles and concept of nozzles
3. Discuss the selection criterion of various solid propellants.
4. Examine the configuration of liquid propellants and feeding.
5. illustrate the combustion instability in rockets and compare the principles of electrical rockets